Imperial College London

AERO96005 - GROUP DESIGN PROJECT

Individual report

Wholly Reusable Launch System (Recovery)

Project Coordinator

Student: Cyril Gliner
CID: 01331358

Project Supervisor: Dr. E. Levis
Team Supervisor: Dr. O. Buxton
Submission Date: 15/06/2020

Department of Aeronautics
South Kensington Campus
Imperial College London
London SW7 2AZ

Abstract

This report presents the work I did as one of the Recovery Team Project Coordinators for the Wholly Reusable Launch Systems Group Design Project (WRR GDP). The outcome of the project is *SynchroCapture*, an unmanned synchropter capable of recovering falling rocket stages. My tasks as part of this project encompassed team management and leadership, as well as overseeing the design and definition of the recovery mission. I also produced a business case for the parent helicopter company, *Recoveroo*.

On the management side, we implemented a strategy whereby we would base our decision on Engineering considerations and common sense. We fully delegated the technical work, entrusting the team members to execute their independent tasks. The conclusion of the business case was that thanks to the recovery of the first and upper stage, the cost per launch could be decreased by up to 55%. In addition, it was found that over a period of 10 years, the *Recoveroo* company can hope to sell around 50 helicopters, mainly by expanding the business into different market. Initial estimates found a possible revenue of around 405 million $ over 10 years, with a profit of roughly 6 million $ per annum.

Acknowledgements

I would like to thank the supervisors that oversaw this project, Dr. E. Levis, Dr. F. Montomoli, Dr Z. Khodaei Dr. M. Ribera Vincent & Dr. A. Wynn. A special thanks goes to Dr O. Buxton, for helping and advising the coordination team throughout the whole project.

I would also like to thank our whole GDP team, for their hard work and dedication during these 5 weeks. Every single person pulled their weight and made this project fun and enjoyable for all of us.

WRR Team Members

Table 1: Team Members and Roles.

Name	Team Name	Team Code
Alessandro Bullitta Yassine Ghenima Cyril Gliner	Project co-ordination	WRR01
Chi Wing Cheng Alessandro Rossi Polvara Tin Hang Un Le Yin	Aerodynamics	WRR02
Anton Le Tejasva Malhotra **Atri Sharma** Alastair Wood	Structures & CAD	WRR03
Omar Arangath Thomas Cross Zuzanna Janusz **Sam Jenner** Jo Tan Tom Walker	Systems & Propulsion	WRR04
Filip Bastien Ananya Dubey Kai Hon Andrew Ng India Tavares	Flight Mechanics, Performance & Control	WRR05
Tom Alston **Hakan Serpen** Tze Leung Wong	Flight Simulation & Testing	WRR06

†Sub-team leads are shown in **bold**

Summary of Design

Table 2: Summary of Key the Specifications for SynchroCapture

Parameter	Value (Metric)	Value (Imperial)
Global Mass & Power		
Weight with Payload (MTOW)	2907 kg	6414 lb
Empty Weight, Helicopter	1229 kg	2709 lb
Fuel Capacity	278 kg	619 lb
Payload Mass	1400 kg	3086 lb
Power Required (Ceiling, ISA)	1193.12 kW	1600 hp
Rotor (For one Rotor)		
Number of Blades	2	2
Blade Mass (each per blade)	40 kg	88 lb
Rotor Diameter	8.05 m	26.4 ft
Rotor Aspect Ratio (AR)	14	14
Rotor Solidity	0.0909	0.0909
Rotor Disk Loading	45.9 kg/m^2	9.4 lb/ft^2
Rotor Blade Loading	503 kg/m^2	103 lb/ft^2
Rotor Hover Tip Speed	205.4 m/s	673.9 ft/s ($\approx$0.65 Mach at 20000ft)
Rotor Cruise Tip Speed	268.529 m/s	881 ft/s
Rotational Speeds		
Engine Output Shaft	1256.6 rad/s	12000 RPM
Rotor Speed	51.1 rad/s	487.5 RPM
Travel Speeds		
Cruise Speed	185 km/h, 51 m/s	99 kts
Cruise Speed (Loaded)	100 km/h, 27 m/s	54 kts
Max Speed (Tip Mach = 0.85)	225 km/h, 62 m/s	121 kts
Never Exceed Speed, v_{ne} (Tip Mach = 0.92)	243 km/h, 67 m/s	131 kts
Speed (Supersonic Flow, Tip Mach = 1)	264 km/h, 73 m/s	143 kts
Capture Velocity	144 km/h, 40 m/s	77 kts
Timings		
Total Mission Time	1 hr 49 mins	1 hr 49 mins

Figure 1: CAD rendering of final design

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1 Introduction

This report summarizes the work I did as one of the Project Coordinator for the Wholly Reusable Launch System Group Design Project. My primary focus was to coordinate the design of the Recovery Systems, alongside two of my peers. I also overviewed the design and definition of the mission, and built a business case for the recovery vehicle, in order to ensure the economic viability of the project.
This GDP was a bit special for a couple reasons. Firstly, this project essentially consisted of two projects combined into one: a team designing the launch system and a team designing the recovery systems. Thus for this project to be successful not only did the communication and collaboration within our own team need to be excellent it also had to be of a high standard with the Launch team. Secondly, the very open-ended nature of the brief required that a lot of decisions be taken in a short period of time. Lastly, due to the COVID-19 pandemic, this project was run completely remotely - not an easy task for a truly multidisciplinary design effort requiring extremely high level of collaboration between every single team member.

2 Leadership & Management

This section describes how along with my two fellow coordinators, Yassine Ghenima and Alessandro Bullitta, I attempted to organize our team to create a productive and enjoyable work environment, in order to produce the best design possible in the allocated five weeks.

2.1 Overall Management approach

The day before the project begun, Yassine, Alessandro and I had a meeting to discuss the management style and approach we wanted to implement. The first decision we made was on the task division between ourselves. We decided that all three of us would work together on the coordination aspect of the project, rather than choosing one of us to do that. This would allow us to run ideas past each other and bring multiple point of views to the table, which we deemed essential when making important engineering and management decision. Furthermore, as all three of us inherently have slightly different approaches to people and problems, this allowed the team a choice of who to communicate with when they had questions or issues.
We then laid out the foundation to the management style and work environment we wanted to create. We quickly established that we wanted to give the team as much technical freedom as possible. We wanted to intervene in the technical and creative decision as little as possible, trusting our team as much as possible. We wanted to create an environment where everyone felt valued and involved and not frustrated because we were imposing our decisions.
Our job would consist in structuring the work to be done at the beginning of each week, and then guide the different teams by keeping a 'big-picture' overview and intervene in the decision-making process only if there was a conflict and/or disagreement between different parties. This does not mean that once we set the tasks to be done, we let the team handle all the work. We stayed aware at all point about all the design decision being taken to make sure the overall design philosophy was coherent, and also to make sure that the relevant teams were made aware of major decisions or changes.
Due to the relative small size of this project, and the fact that we knew very well most of the other team members, we did not see to rely on management theories, which in our opinion are more suited for larger teams and projects or production lines. Instead, we chose a management style and decisions on engineering considerations and good practice.

We then proceeded to choose the means of communication we would be using. Because this project was run remotely this decision was quite important. All three of us agreed that a centralized approach with as much as possible done in a single location was best, and not use many different platforms which will

ultimately cause problems with lost data, compatibility issues etc. As such we elected to use Microsoft Teams for everything: meetings, chat platform, file sharing and task planification. Teams offered all of this in one place so was an ideal choice.
We created a general channel for team-wide communications and update, as well as individual sub-team channels. As will be explored later, we sometimes re-organised the team to work on specific tasks. When this was the case we would just create a new channel to facilitate the communication between this newly formed sub-team.
From past experiences, we knew that online meetings with a lot of people tend to be inefficient and unproductive. As such, we decided to minimize the number of meetings as much as possible, and always keep them small, only inviting people necessary to the discussion. With this in mind, we decided against daily or even weekly team wide meetings, which we judged would only be a loss of everyone's time. Rather we decided to have each team choose a sub-team coordinator, with whom we were in frequent contact during the whole project. They would keep us updated daily by message as to what their team have been doing and what their next steps were. This allowed us to stay up to date with everything going on, without large meetings. We also encouraged individual people to communicate between each other and with us via message, to ensure a good flow of information.

2.2 Design Philosophy

The ultimate aim of the project was to come up with a realistic and functional yet innovative and ambitious first iteration of the design. This might seem somewhat contradictory at first, but this is because we not only strived for an interesting and optimized design on paper, we wanted our design to be feasible and be able to move on to the production phase should we want to. Throughout the project we encouraged the use of novel techniques, such as using generative design to size the fuselage, but remained realistic and steered the team away from decisions such as a fully autonomous and electric helicopter.

2.3 Pre-conceptual design phase & Vehicle Selection

Week 1 of the project was devoted to the pre-conceptual design phase, which in our case corresponds to the vehicle selection. We short-listed three vehicle configurations that could be used: a fixed-wing aircraft, a helicopter, or a hybrid VTOL vehicle. Instead of sticking to the provided team division by discipline, we decided to split the whole team into three sub-teams, one for each vehicle. Each team would investigate their respective vehicle during the week, and on Friday we would compare the different solution proposed and make a choice. We organised the three teams to make sure that each discipline was equally represented to ensure that all aspect of the vehicles are considered. One coordinator was designated to take the lead of each concept vehicle - I lead the helicopter concept.

2.3.1 Helicopter Concept

I was leading a team of seven to come up with proposition for a helicopter design capable of recovering falling rocket stages, as well as a feasibility analysis. The helicopter concept proved challenging as none of us had ever worked with helicopters before and we did not know much about them. As such, during the first team meeting I told everyone to simply research helicopters first, to familiarise themselves with the technology. From there, I split the team into two; it was decided that three people would try to estimate rough sizings and costs for a helicopter capable of fulfilling our mission based existing helicopters. The rest of the team explored more innovative helicopter designs that we could potentially investigate. This team was also responsible for coming up with different ideas for the capture mechanism that we could use.
Towards the end of the week, I had the whole team back working together on a summary of all the work that had been done, focusing on clearly summarizing the pros and cons of using a helicopter,

particularly focusing on a few specific metrics (cost, feasibility, range, catch speed, manoeuvrability and payload capacity).

2.3.2 Final vehicle selection

Once all three teams had compiled a summary of the pros and cons of each design, we performed a trade-off study to evaluate which would be the best option.
We wanted to compare each of the vehicle based on the criterion mentioned above. To do so, we designed a weighting matrix to compute the relative importance we wanted to give to each criteria.
The weight matrix contained all the key factors that we based the comparison upon. The relative importance of each factor was determined by comparing each factor to all the others and assigned a score from 1 to 10, 1 being the less important, 10 being more important. We then took the normalized mean of each factor, and that gave us a ranking for the least to most important factor. The results and the weight matrix can be can be seen in Figure 9 and 10 in the Appendix.
At the time when we did this analysis, we had not yet received the final confirmation from the Launch team for the uncertainty in the recovery location's estimate. This was however a determining factor for the vehicle selection, as a helicopter travels relatively slowly compared to the other options and would not be able to accommodate an uncertainty greater than 50 km. At this point, we had an important decision to make make. We could wait a few days for the launch team to get a final value, although that meant delaying our work for a few days, which we could not afford, given the tight timeframe. Instead, we decided to perform a risk analysis, to take into account that there was a chance albeit small that the helicopter might not be able to perform the recovery altogether. We also included the feasibility of the design and likelihood of success of the catch manoeuvre in this risk analysis. A score from 1-10 was given to each parameter, 1 being low risk and 10 high risk. The scores were then added together and normalized, to give a risk factor. This risk analysis can be seen in the Appendix, figure 11. Overall this gave us a tied score between the VTOL and helicopter, with the fixed-wing being deemed slightly riskier, due to its significantly lower likelihood of successful catch.
To make the final choice, we looked at the three most important factors (as determined by the weighting matrix), costs, feasibility of the design and likelihood of success. Based on this, the helicopter appeared to be the best suited for the mission. It is significantly cheaper both to develop and operate compared to the others, and the catch manoeuvre was deemed the most likely to succeed. The only issue was the operational range and small dash speed, which could be a problem in case of a large range uncertainty. We had included this in the risk analysis however. Furthermore, the launch team had guaranteed an within two days - which meant that because this remains a pen and paper exercise and not a real industrial project, we could always do a U-turn two days later (although clearly not ideal), if we realized the helicopter was inadequate.

2.4 Team organisation for the rest of the project

Now that we had decided to use a helicopter, we could move on to the conceptual and preliminary design phases. The first thing to do was the baseline configuration. Because the team organisation of the first week, paring different members in task-groups rather than by disciplines worked really well, we decided to continue with this structural organisation.
We would create the new tasks-groups in collaboration with the whole team, asking them for input. We made sure to always have some team member from all the relevant discipline involved in each tasks, to make sure no aspect was getting neglected. As an example, for the fuselage design, the team was made up of people from Aero, Structures and Systems, all working together, to come up with a solution that would satisfy all the constraints, and not be something very good aerodynamically but impossible to manufacture. As coordinators we would over watch the design to make sure that costs were also taken into account, and that nothing major was being neglected. We quickly realised that having someone with

the bigger picture in mind, rather than focusing on small technical detail is extremely important and can allow at times to catch flagrant errors that can be overlooked by the team when they are focusing on the minute technical details. As an example of this, the performance team computed that the ceiling of the helicopter with the payload was upwards of 20000ft. I knew however that the engine had been sized for a ceiling with payload of approximately 10000ft. After asking for a double check, it was quickly found that there was a slight error in the power lapse model. Having someone with an high-level overview of the entire project double checking the result, but still giving the team the technical freedom and authority proved to be a very efficient technique.

There were two occasions where we had to mediate the discussions and intervene to make an engineering decision. The first one came early on in the baseline configuration phase. The aerodynamics team wanted to include a tiltable back rotor, to improve forward speed as well as manoeuvrability. The Flight Mechanics team were however opposed to this idea, as it would make their modelling even harder than it already was, and they judged they would not be able to model it in the given timeframe. To solve this problem, we held a meeting with the rotor sizing team, to come to a solution. It was established that whilst the tail rotor would be a nice addition and would undoubtedly improve performance, it was not strictly necessary. As such, we decided to trade some performance for a more realistic design that we could actually model. This was in line with our innovative but *realistic* design philosophy.
The second time we had to intervene in a major decision was when the aerodynamics team reported a mistake in their initial calculation which meant we needed bigger blades. Again here we were faced with two choices: keep the current blade at the cost of increasing the disk loading, resulting in a lower hover efficiency, or change the blade, which would mean everything (from the engine to the fuselage) would have to be resized. Because this was the end of week 3 already, after weighing up both options, we decided to go for the former. It would not have been reasonable to resize everything in the time left, and could have potentially led to more problems down the line due to rushed calculations. Instead, we accepted the decreased efficiency, as our ultimate goal was having a working design at the end of the project. This problem could be fixed in a second design iteration.

2.5 Coordination with the Launch Team

As mentioned previously, getting the coordination with the launch team right would be essential to have an overall coherent and good design. This would prove a bit trickier than we would have hoped. Alessandro [3] and Yassine [4] speak in their report about the problems we encountered regarding the mass of the payload. I mainly worked with them on the mission definition, for which we also had some problems. As explained later in the report, one of the decision we had to make was whether we would use 1 or 2 boats. To make this decision we needed input on the re-entry trajectory. However, after waiting almost 2 weeks we still had no answers, and the coordinators seemed to be contradicting their own subteam, which created some uncertainties. Contrary to us, the coordination team in the launch group elected to be heavily involved in the decision making and in the technical work. From what we understood this caused some tensions, as the coordinators were in disagreement with their technical team, which led to inevitable disagreements, frustration and delays, which in turn affected our work. To remedy this problem, we elected to deal directly with each sub-teams directly, as opposed to ask group-wide questions, to which we would get contradicting answers. Once this was in place, things improved and we started getting numbers and results. The final verdict for the boat still caused a lot of disagreements within their team, and we decided to put it to a vote in the end, as the discussions were not going anywhere.
Whilst the above paragraph seems to paint quite a negative picture, it is only because the negative tend to stand out more than the positive. Overall the work I have done on the mission planning and on the recovery manoeuvre, along with the Systems, Trajectory and Ground Systems team was very positive and fruitful, and lead to what we think, is a very elegant recovery solution.

2.6 Reflection on management

Overall we were mostly pleased with how the project went, and thought that the structure and management style we implemented were successful. There were some problems at times, both internal to our team and with the launch team, but we tried to iron these out as we went. The feedback we were getting from our team during the project was in the main part positive.
The main point for improvement would be how we collaborated with the launch team. Because of the asymmetric relationship between us, where we depended on team much more than they depended on us caused some tension from the get-go and we never fully managed worked in sync. They were using completely different platforms than we were, and our timelines were not aligned. It is our opinion that a different organisation of the management team would have been beneficial. Instead of splitting it with three distinct managers for each project like it was, it maybe would have been better to be a single team of six. This would have allowed to treat this project as one large project rather than two small one. This would have ensured that all decisions were taken with the overall big-picture of both the recovery and launch systems in mind, and the common aligned goals, not simply the rocket or the helicopter.

3 Mission definition

Throughout the project I was responsible for the design of the helicopter mission profile, which served as a basis for the work done by all the subteams, as well as the design of overall mission definition (in collaboration with the Launch team).

3.1 Helicopter Mission Profile

The design of the helicopter mission profile started at the very beginning of the conceptual phase, and was progressively refined as the weeks went by, with numbers and values slowly getting updated. The final mission profile is detailed below, and illustrated in Figure 12.

0-1 Engine warm-up + takeoff - 5 mi

1-2 Climb and cruise at 185km/h to recovery altitude and site - 75 km, 20000ft.

2-3 Lookout for the payload. Hover 10 min.

3-4 Perform the recovery manoeuvre (constant 15 deg descent from 20000ft to 3000ft, at 35 m/s)

4-5 Cruise back with the payload attached at 100km/h – 75 km.

5-6 Safely deposit payload - 5 min

6-7 Land - 5 min

3.2 Overall mission definition

Designing the overall mission definition consists in defining the entire recovery mission. Essentially, it consists in answering the following two questions. What will we be recovered? & How will the items be recovered by the helicopter? The latter question also includes considerations such as where will the helicopter operate from and how many helicopters will be required?
These questions are addressed in the following subsections.

3.2.1 Operating from Land or Boat?

One of the first decision we investigated was whether we will be operating from land, or from a boat. The launch team chose to do all the launches from the Kourou launch site in French Guyana [5]. This means that the rocket will have a trajectory over the Atlantic ocean and the recovery sites will be a few

hundred nautical miles away from the shore. This naturally suggests operating from a boat. However, after doing some research on boat prices, we soon realized that boats that could be used as the recovery vessel extremely expensive. Based on some research done by Hakan [6], renting a single boat would cost around 95000$ per mission and per boat. We thus looked at buying a boat instead. Based on similar sized boats and vessels, it was estimated that buying a boat of this size and capabilities would cost around 8 milion $. Using an online operating cost calculator [7], it was estimated that over the 10 years of planned service an extra 4 million $ should be added in operating costs. Assuming an average of 40 missions a year that gave the approximate boat costs as 30000$ per mission if we bought a boat, which is turned out to be much less than renting one. This was still a very steep price-point however. Alessandro [3] estimated the helicopter price per hour to be around 4642$ (based on 80 flight hours per year). Assuming 1.5 hours of helicopter flight time per mission, this meant that around 60% of the recovery costs were due to the boat.
Based on this result, we thus decided to explore alternative 'boatless' ideas. We proposed to the launch team adding some steering devices to the stages to be able to perform the recovery closer to the shore so that we could operate from the land directly, getting rid of the need for a boat. This was however quickly rejected by the launch team, mainly on grounds of safety. They deemed it unsafe to perform the recovery close to the shore, where there are potentially commercial and private boats. Operating from oil rigs was also explored as an alternative, but was quickly abandoned, mainly because there are no guarantees that an oil rig would be in proximity of the recovery site (which will depend on the customer orbit).
Although the cost of boat seems large, one must remember the bigger picture and the entire mission, not just the recovery segment. When compared to the overall mission cost, including the rocket ($\sim$ 1.2 million $ per launch), the price of a boat still remains acceptable.
Based on all of the above, it was decided that we will operate from a boat. A CAD rendering of what the boat could look like in shown in Figure 13.

3.2.2 What to recover?

The primary aim of this project was to *completely* recover the orbital launch system. The upper stage had to be recovered aerially, whilst the first stage could either be recovered aerially or landed using propulsive methods (much like spaceX does [8]). Because the rocket had to be wholly reusable, should the payload fairings be jettisoned, they should also be recovered.
The first decision to be made was whether the first stage will be aerially recovered or not.
It was found during the pre-conceptual design phase that a helicopter could be sized relatively easily to carry payloads of up to 4 tons. After some rough initial sizing, the launch team quickly established that the empty weight of the first stage (the heaviest payload to be potentially recovered) would stay well below 4 tons. Therefore, as far as our team was concerned, aerially recovering the first stage was doable.
The launch team also looked into this question. They estimated that landing the rocket would roughly double their development and operation costs as compared to using an aerial recovery method [9], due to the complexity of the re-entry systems and to the added mass of the extra engines and fuel required.
Because the rocket development cost is estimated to be around 150 million $ [10], we did not even have to estimate the cost savings that could be obtained by designing a helicopter capable of carrying only the lighter second stage and fairings, as it would be much smaller than the extra rocket costs due to the propulsive landing. It was thus decided that the first stage was to be recovered aerially.
We then shifted the focus onto the question of whether the fairings should be jettisoned and then recovered, or kept on the upper stage scraping the need for an extra recovery, at the cost of bringing more mass into orbit. This was not as clear cut as with the first stage, so it was decided to perform a more detailed analysis of the costs and feasibility analysis for both options.
For the recovery side of things, we decided to look at the cost of the two options mainly, as we deemed both options equally feasible.

To recover the fairings, two helicopter and one boat would be needed as the two fairings will need to be recovered at almost the same time. It is however fair to assume the recovery sites to be close to each other, justifying the need for only 1 boat. Using the same prices as defined above in Section 3.2.1, the extra costs of recovering the fairings were calculated to amount to a a non-negligible 52.76 thousand dollar cost per mission.
We passed on these values to the launch team who also conducted a cost and feasibility analysis. In short, their analysis found the following results if the fairings were kept on the upper stage.

+ Less things to design for re-entry; no need for parafoils or cold gas thrusters to be added to the fairings; Cheaper recovery; Inside of upper stage protected from the atmosphere during re-entry.
- Keeping the fairings would cause a 3 tons increase in wet mass. Whilst this is significant, it is largely out-weighted by the lower recovery costs and much simpler design. More details on the specifics reasons can be found in [9].

In summary, it was decided to aerially recover the first and upper stage, and to keep the fairings attached to the upper stage.

3.2.3 Recovery strategy and Vehicle Management

Once we had established that we had two stages to recover, the next step was to design determine how these recovery would take place.
Without any sort of planned re-entry trajectory, the upper stage would fall far away downrange from the first stage. This would thus call for 2 boats and 2 helicopters; each helicopter and boat recovering one stage. Whilst satisfactory, we decided to innovate and along with the launch team we tried to work out a way to use only 1 boat and 1 helicopter to recover both stages. The idea was to have the upper stage perform an extra orbit before re-entering the atmosphere. This would allow the possibility to time the de-orbit burn in such a way to have the recovery site close to the first stage recovery site, only a few hours later. Due to the earth's procession, the upper stage will not re-enter at the exact same location as the first stage, but hopefully sufficiently close. To make sure this was feasible, we studied whether the helicopter could cover the distance between the two recovery sites in the given timeframe, and if this idea had any economical merit.

To do so I used Matlab® to estimate what would be the maximum distance that could be covered within a given time, by moving both the boat and the helicopter in an optimised manner. The problem had the following conditions:

- Both payloads could fall anywhere within a 50 km radius circle (defined as the recovery area uncertainty).
- Without any payload, the helicopter can travel up to 225 km/h. With the payload, the helicopter can travel up to 100 km/h. The boat can travel at 27 km/h (15 kts).
- The helicopter and boat initially be positioned anywhere, and move in any direction.

The aim is to perform both recoveries in the most efficient manner as possible. Here we defined the most efficient solution as the one allowing for the greater distance between the two recovery areas in a given amount of time.
The mission profile goes as follows: The helicopter travels from its initial position to the recovery site (anywhere in the circle). It then performs the recovery of the first stage, travel back to the boat, deposit the first stage, refuel and then cruise straight to the upper stage recovery area to attempt the second recovery.

The only thing that we can control is the starting position of the boat and helicopter (i.e before the exact recovery site is known), and then their movements, and finding the optimal position is the goal of this exercise.
This is a highly complex non-linear optimization problem, as it involves circles and moving parts. For the sake of time, I decided to use a 'brute-force' method, rather than a more elegant optimisation method. The idea was to find the starting position and movement of boat + helicopter that had the *best worst case scenario*. The worst case scenario is defined as the scenario where the payloads falls in the location that would take the longest to perform the recovery, given that boat+ helicopter position. For each boat/helicopter pair position, the worst case scenario was computed, and compared to the other worst case scenarios for other pairs. The optimised solution was then the pair that had the least worst case scenario.
The metric used to compare the different solution was chosen to be time for recovery. The longer the time for recovery, the worst. The overall time computed consisted of 3 segments, taken directly from the mission profile, available in section 3.1. All the fixed time segments (takeoff, landing, hover) were not added for the comparison as they are equal for all cases. Note that the hover time at the recovery site was scrapped, as we are optimizing for speed and efficiency. During the recovery, which lasts 12 minutes (as given by [5]) the parafoil is assumed to travel up to 20 km in a chosen direction. During these 12 minutes, the boat was allowed to move to any location within a 5.5 km radius (the distance it can cover in 12 minutes). To optimize distance covered, the boat will be moving during the landing and refuelling stages as well.

3.2.4 Result of the optimisation

The result of the optimisation described above can be seen in Figure 2. The icons mark the optimal initial position of the boat and helicopter. The parachute represents the worst case scenarios for the recovery site for these initial positions.
To my somewhat surprise, the ideal combination is not simply starting with the helicopter at the centre of the recovery area. This is a rather counter-intuitive result, partly due to the speed difference of the helicopter with and without payload, but also due to the fact that the boat can move.
As a sanity check to make sure the algorithm was working, I considered the extreme case where the helicopter is extremely slow with the payload ($V_{\text{cruise}} \gg V_{\text{payload}}$). The result for this simulation are shown in Figure 3. Essentially in this configuration the only leg that matters is travelling from the payload to the boat, as this one will take by far the longest. Placing the boat anywhere else but in the centre would mean that that this leg could potentially be greater than 50 km, increasing the time. In this case the same reasoning applies to the helicopter. Therefore the optimal solution is placing both the boat and helicopter in the centre of the recovery area. This solution is much more intuitive, and the algorithm seems to work well.

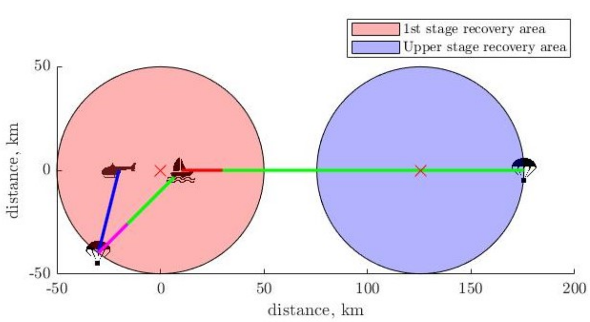

Figure 2: Result of the optimisation algorithm for the vehicle positioning

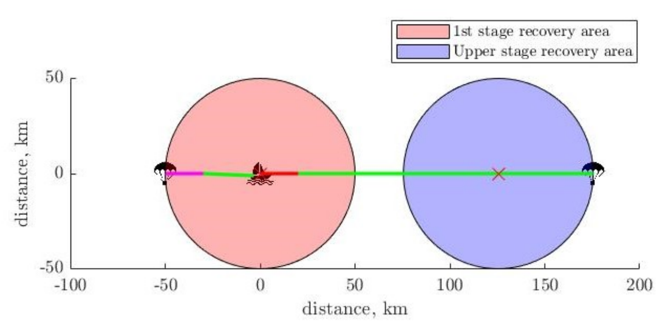

Figure 3: Result optimisation algorithm with $V_{\text{paylaod}} \ll V_{\text{cruise}}$

The launch team gave a preliminary estimate that the upper stage would land around 1.5 hours later than the first stage, around 50 km away. According to the optimisation above, in 1.5 hours the helicopter would be able to recover the 2 stages as long as they fall within 95 km of each other. Thus given those numbers, it is possible to use only 1 boat and 1 helicopter to retrieve the rocket in its entirety.

There was however a problem with this idea. For orbits with a somewhat large orbital inclination ($i \gtrsim 30$ deg), the upper stage would require a large translation during its re-entry to land close to the first stage. In terms of trajectory and fuel required, the launch trajectory team [11] seem to suggest that this was not a problem at all. However because the rocket is launching from Kourou in French Guyana, this trajectory would have the upper stage pass over the Caribbean islands. The FAA regulations [12] stipulate that the risk to the public if something was to go wrong must be $< 10^{-7}$. This would be hard to guarantee over the Caribbeans, and would essentially require a very large exclusion zone. A zone of what this exclusion zone could be is sketched in figure 4. This is a *rough* estimation, not based on any quantitative numbers. Enforcing this exclusion zone would be very difficult and costly. Furthermore, due to the abundance of airports in the region, as shown in figure 5, many flights would possibly need to be cancelled. These reasons make the single boat recovery almost impossible for high inclination orbits. Instead of just scrapping the 1 boat idea however, a compromise was found. Depending on the inclination of the orbit, 1 or 2 boats and helicopters will be used. If $\lesssim 30$ deg, then a single boat and helicopter will perform the recovery, with the upper stage doing an extra orbit before re-entry. However if $i \gtrsim 30$ deg, then each helicopter and boat would only recover 1 stage. There would be no need here for the upper stage to perform an extra orbit.

3.2.5 Summary of Mission definition

In summary:

- Both first and upper stage to be recovered aerially, with the fairings kept on the upper stage
- If $i < 30$ deg, one boat and one helicopter to perform all the recovery. Else, two boats and two helicopters needed.

4 Business side of the project

As with any project, coming up with a great and efficient design is one thing, but it must also be economically viable. If it is not, even a flawless design might never see the light of day. In this succinct business case, we will explore the different mission costs, as perform a market analysis to estimate how many helicopters we can hope to sell to estimate the potential profits that our helicopter company could make.

Figure 4: Estimation of the exclusion zone required for re-entry from polar orbit

Figure 5: Mission profile for a single stage recovery for re-entry from polar orbit

Table 3: Recovery costs for the different mission types

	Cost over 10 years (mil $)	Cost per mission (thousand $)
2 boats, 2 helicopters	40.44	77.76
1 boat, 1 helicopter	20.92	40.23

4.1 Some quick numbers

The launch team [9] are targeting 1 launch per week at < 1.2 million $ per launch. The aim is to be operational within 5-10 years.
To compute the recovery costs, the two possible mission scenarii were considered (small or large inclination orbit). As previosuly mentionned, Alessandro [3] estimated acquisition price and operating costs of the helicopter, which he found to be as follows. Acquisition price: 3.52 million $, Direct Operating Cost (DOC): 892 $/hr. Indirect Operating Cost (IOC): 300,640 $/yr, giving a total flight hour price of 4642 $ assuming 80 hours of flying per year.
Assuming 1 launch per week over 10 years, that if the helicopter is recovering only one stage it flies for 1.5 hour, and for three hours if recovering both stages and the costs of the boat are (as defined in the previous section) 12 million $ over 10 years, we can calculate the overall mission costs for both scenarii. These are show in Table 3.

4.2 Cost saving thanks to the recovery

As seen above, the launch team estimates a launch at around 1.2 million $ [10]. They also estimate that if the vehicle was expandable, then each launch would cost around 2.85 million $.
It was decided that our two companies, *Recoveroo* and *SynchroSystems* would form a joint venture. *Recoveroo* would sell 'recoveries' as services. Because of the joint venture, a healthy 10% profit margin was agreed upon for Recoveroo. This means that for a recovery requiring two boats and two helicopters, *Recoveroo* would charge $1.1 \times 77.76 = 85.536$ thousand $ for the recovery of the rocket. If we add this to the initial 1.2 million $, and compare it with the case where the rocket is not recovered, we can see that the recovery of the rocket allows for a staggering 55% reduction in the launch price!

4.3 Market analysis

The primary aim of our helicopter is to recover rockets. To perform this task for the next 10 years, realistically only around 4-5 helicopters will need to be produced. Assuming we also convince Rocket Lab to buy our helicopter instead of the one they are currently using, we can maybe aim to build and sell around 10. Due to the very high development costs associated with the design of a novel helicopter from scratch, building only around 10 units would result in a prices much higher than the ones presented above.
As such, we decided to explore the idea of expanding our business outside of the rocketry world. The goal is to also be able to sell our helicopter to perform other tasks.
To this end, instead of creating a single company that would incorporate both the rocket and recovery part, we decided to go for 2 separate companies, which will greatly facilitate our expansion into other industries.
To determine how many helicopters we could sell, we decided to perform a short market analysis.
We first established which markets we could sell our helicopter in. The main strengths and characteristics of *SynchroCapture* are its High Power/Weight ratio thanks to its intermeshing blade configuration, which makes it very suitable to cargo transport, and the fact that it is unmanned.
After some research on the uses of helicopters, we shortlisted the following markets that *SynchroCapture* could enter:

- Aerial fire fighting. The high-lifting capacity means that*SynchroCapture* could carry up to 2.5 tons [13] of water to help fight fires. This is well within the amount that current firefighting helicopter such as the Bell 205 [14] can carry. The fact that it is unmanned however gives *SynchroCapture* a great safety advantage over the other traditional manned helicopters.

- Aerial Crane. Thanks to the high-lifting intermeshing rotors, *SynchroCapture* can carry way more payload than a similar sized helicopter. Whilst most helicopters have an empty weight to payload ratio of around 0.7 [15], *SynchroCapture* is estimated to have a staggering 1.5.

- Military & Defence. Again, thanks to the unmanned and high lifting characteristics of *Synchro-Capture*, it could prove a very valuable vehicle for any military in the world, as it would allow to deliver large amounts of cargo in remote and possibly dangerous locations with a reduced risk.

Using a market reserach done by Statista [1] in 2019, and the above information, we then tried to estimate how many helicopters we could reasonably sell. Figure 6 shows the total number of helicopters sold per year. In recent years the trend seems to have settled around 1100 helicopters sold. Note that the Leonardo data is missing for 2019 which explains the significant drop. Figure 7 gives the Market share of the leading helicopter manufacturers worldwide in 2019. As expected in the aerospace industry, the market is ultra-dominated by a few key players. This is due to a number of factor ranging from reputation, a proven track-record, and the large advantages to be gained from economies of scale and past experience. This will make entering the market as a brand new company challenging.

Figure 8 illustrates the projected uses for new commercial helicopters from 2020-2024. The use cases of *SynchroCopter* fall under the General/utility/other segment, which is expected to represent 33% of the sales in the years to come. Unsurprisingly not much could be obtained concerning military helicopters. The Statista report did mention however that the military market size represents approximately half of the commercial market size. For the purpose of this business case, we will make the (rough) assumption that half the market size can be approximated to mean half the number of helicopter shipped. Whilst not really representative, given the lack of military data this was the best we could do in the timeframe.

Combing all of this data together, if we aim to capture around 1% of all the markets we join, over 10 years, we would expect to sell around 40 helicopters. Adding this to the 10 helicopters sold for rocket recapture, we would in total be looking to sell around 50 helicopters. Whilst this will not be quite enough to classify *SyncroCapture* as a mass produced helicopter, it should be enough to reduce the absorb the development costs and allow the acquisition price to be the targeted 3.52 million $.

4.4 Profit estimation

We thought it would be interesting to come up with a rough estimate of how much profit we could get from *SynchroCapture*. After discussing it with the launch team, they wanted us to operate as a service, selling recovery missions rather than simply selling the helicopters. Because we would operate as a joint venture, we agreed (after some tense negotiations of course), on a 10% profit margin. The prices we gave them were calculated for 10 years of operation (see Table 3). Given a 10% profit, and assuming 10 helicopters are to be used for rocket recovering, that would bring our profit to around 20 million $ over 10 years. Add to this another 40 helicopters sold for other purposes (at a 30 % margin, not partners any more!), this brings us to a total profit of 62 million $. The total revenue would be around 405 million $.

Note that these are only the revenues of the Recoveroo company, considering only the sale of the helicopters. All the potential revenue thanks to support and maintenance are neglected. The revenue and profits of the Launch company are not included.

Also note that these numbers are very rough estimates and were only calculated for added fun.

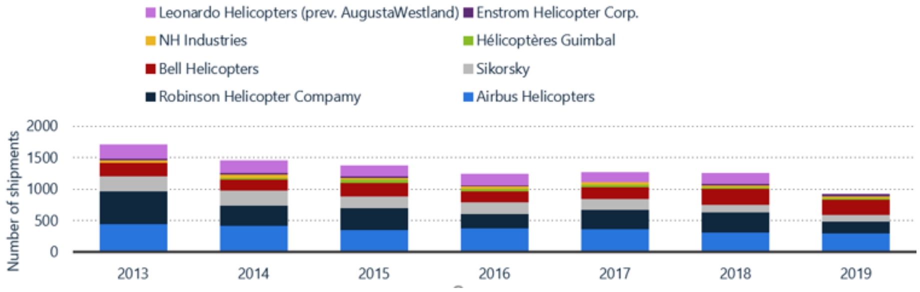

Figure 6: Worldwide rotorcraft shipments by manufacturer 2013-2019 [1]

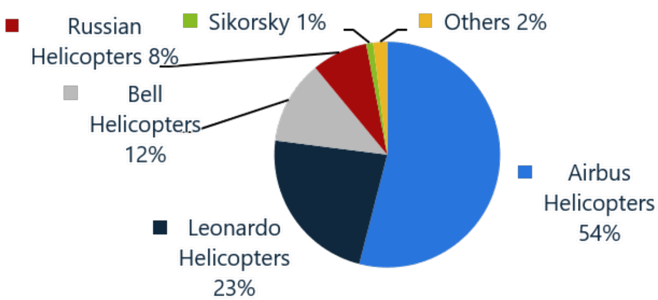

Figure 7: Global helicopter market share by manufacturer 2019 [1]

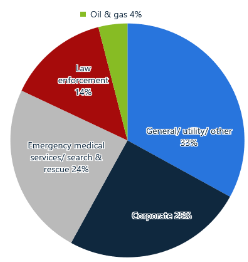

Figure 8: Projected global commercial helicopter deliveries from 2020 to 2024, by use [1]

5 Conclusion

Through this project, I really got to appreciate how challenging managing a team working on a truly multidisciplinary project is. Because of the very short available time (5 weeks), this project taught me the importance of being decisive and to be able to make a lot of decisions quickly based on engineering judgement, even if sometimes only incomplete information was available. Although we ran into a few issues with communication and collaboration, I believe that both were in general mostly very good throughout the project. Furthermore, the approach we put in place worked well. Managing with engineering consideration in mind allowed for great flexibility and agility. The high trust environment we tried to create, with us being minimally involved in the technical work ensured that everyone felt valued and stayed motivated.

One area that we maybe could have improved project-wide is time-management and anticipation. We at times found ourselves with meetings and presentation lasting longer than what they should have. In addition, especially towards the end, everything had to be a little rushed, and we maybe could have anticipated this a bit more and have set deadlines one or two days before.

Overall, we came up with a functional design which, whilst far from perfect, flies and handles well in the simulator, and is very well suited to successfully recover falling rocket stages. It is also worth noting that it is actually a realistic design. In a second iteration, the main area for improvement would be the rotor and blade design, which we found to be the weakspot of the helicopter when testing it in the simulator. As seen in the report, thanks to this recovery method, we estimate that the launch costs could decrease by up to 55%. This will allow for much more affordable space flight, with the ultimate goal of democratizing space flights.

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A Vehicle selection tradeoff study

Parameter	Time for capture	Speed/Range	Cost	Likelihood of success	Pilot workload	Feasibility of design	Maneuvrability and response	Normalized mean
Time for capture	1.00	3.00	0.20	0.14	8.00	0.14	1.00	0.44
Speed/Range	0.33	1.00	0.33	0.20	6.00	0.20	0.50	0.28
Cost	5.00	3.00	1.00	1.00	10.00	1.00	3.00	0.79
Likelihood of success	7.00	5.00	1.00	1.00	10.00	2.00	4.00	0.98
Pilot workload	0.13	0.17	0.10	0.10	1.00	0.10	0.13	0.06
Feasibility of design	7.00	5.00	1.00	0.50	10.00	1.00	6.00	1.00
Maneuvrability and response	1.00	2.00	0.33	0.25	8.00	0.17	1.00	0.42

<1 is low importance, >1 high importance

Figure 9: Weight Matrix used to perform the trade-off study.

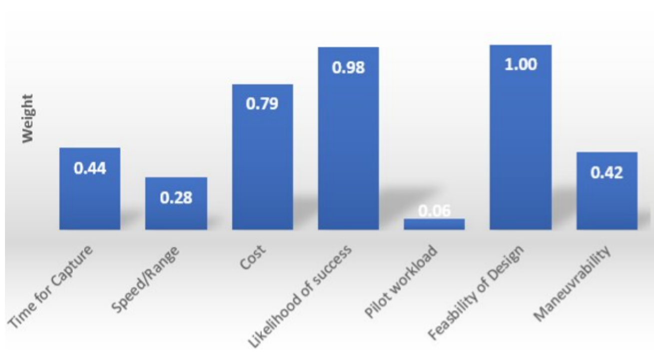

Figure 10: Histogram showing the relative importance of each weight obtained from the weight matrix.

Parameter	Aircraft	VTOL	Helicopter
Time for capture	10.00	2.00	1.00
Speed/Range	1.00	5.00	10.00
Likelihood of success	6.00	4.00	2.00
Feasibility of design	3.00	6.00	4.00
Average	5.00	4.25	4.25

Note: 1 is low risk, 10 high risk

Figure 11: Table used to analyse the risk associated with each vehicle configuration.

B Helicopter Mission Profile

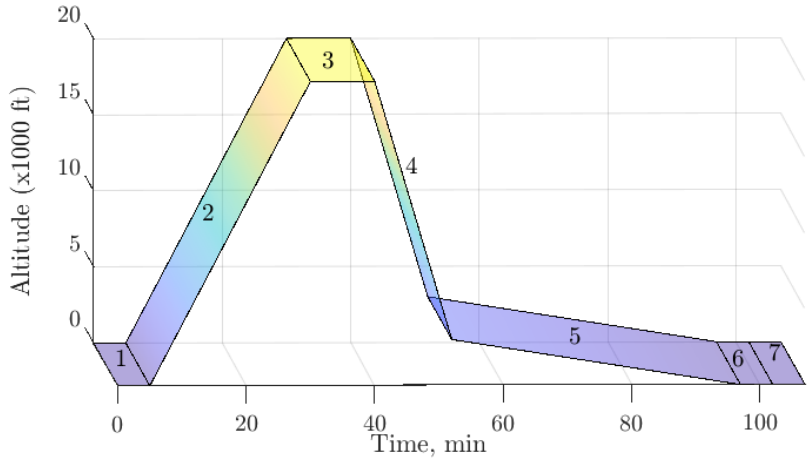

Figure 12: Helicopter Mission profile for single stage recovery

C CAD render of a possible boat design

Figure 13: Rendering of boat that could be used as the operating base [2]